

Title: Geographic and Economic Predictors of Nonprofit Donation Revenue Growth in the United States

Abstract

This study investigates how geographic location and community economic conditions influence the financial performance of nonprofit organizations in the United States. Using data from the Spring 2021 National Survey of Nonprofit Trends and Impacts, the analysis examines donation revenue change as the primary indicator of nonprofit financial health. Contrary to expectations, urban nonprofits do not report significantly higher donation growth compared to their rural counterparts. However, nonprofits operating in low-income ZIP codes experience significantly lower donation growth than those in higher-income areas. These findings highlight the primacy of economic disadvantage, more than geographic location, in shaping fundraising outcomes and suggest a need for targeted support to organizations serving economically distressed communities.

Introduction

Nonprofit organizations demonstrate significant variation in financial health based on geographic location and community economic conditions. Analysis of the Spring 2021 National Survey of Nonprofit Trends and Impacts reveals that while urban-rural differences in donation growth are negligible (4.97% vs. 4.26%, $p = 0.5481$), nonprofits in low-income ZIP codes experience markedly reduced donation growth compared to higher-income areas (3.08% vs. 4.77%, $p = 0.0981$) (Urban Institute, 2021). These findings underscore the primacy of economic context over pure geographic factors in determining financial vulnerability.

Theoretical Framework and Contextual Determinants

Nonprofit organizations' financial sustainability depends on internal management practices and external environmental factors. While prior research emphasizes governance quality and revenue diversification as critical to financial health (Zainon et al., 2011; Varghese & Ajukurian, 2021), contextual variables like community wealth and geographic isolation receive limited attention. Bowman's (2011) foundational work establishes that financial capacity requires maintaining assets at replacement cost through strategic surplus accumulation, a challenge magnified in resource-constrained environments.

Research Questions

Emerging evidence suggests geographic disparities in philanthropic resource availability, with rural nonprofits facing unique constraints. The Urban Institute's (2021) survey data indicates 32% of rural organizations report limited access to major donors compared to 18% of urban counterparts. Simultaneously, community-based development organizations in economically distressed areas show elevated financial stress indicators, including 16% insolvency rates and reduced operating margins (Nesbit et al., 2021). This dual burden of geographic and economic disadvantage creates compound vulnerabilities that existing theoretical models inadequately address (Bridgespan Group, 2019).

Research Aim

This study investigates how geographic location (urban vs. rural) and community economic conditions affect nonprofit financial performance, specifically through changes in donation revenue. Drawing on the financial capacity framework and empirical findings on community

resource disparities, the research aims to test whether location-based and economic-contextual variables predict changes in nonprofit fundraising outcomes.

1. How does the donation revenue growth of nonprofit organizations differ between urban and rural areas?

- **H1:** Nonprofit organizations headquartered in urban areas will report higher donation revenue growth than those in rural areas.

2. Does operating in a low-income community negatively affect a nonprofit's donation revenue growth?

- **H2:** Nonprofit organizations in low-income communities will report lower donation revenue growth than those in higher-income areas.

Research Method

This study utilizes data from the Spring 2021 National Survey of Nonprofit Trends and Impacts, a nationally representative survey of 2,306 operating 501(c)(3) nonprofit organizations conducted by the Urban Institute in collaboration with American University, George Mason University, and Independent Sector (Urban Institute, 2021). This dataset is uniquely suited to examine geographic and economic disparities in financial health, as it includes granular variables on organizational finances, community context, and operational characteristics. The survey employs a stratified sampling design to ensure representation across nonprofit subsectors, organizational sizes, and geographic regions, making it ideal for analyzing systemic inequities in resource allocation.

Sampling Approach

The sample was drawn from the Internal Revenue Service (IRS) Business Master File and stratified by five organizational size categories (annual expenses ranging from <\$100,000 to ≥\$10 million), National Taxonomy of Exempt Entities (NTEE) subsectors, and all 50 U.S. states plus Washington, D.C. (Urban Institute, 2021). Schools, hospitals, and religious congregations were excluded to focus on community-serving nonprofits. To address the underrepresentation of small and rural organizations, the design incorporated oversampling in these groups, with 32% of the final sample representing nonprofits in non-urban areas (Nesbit et al., 2021).

Weighting and Representativeness

Survey weights (`weight_complete_partials`) were applied to correct for unequal selection probabilities and non-response bias, ensuring the sample reflects the national population of 501(c)(3) organizations with annual expenses ≥\$50,000 (Urban Institute, 2021). The weighting scheme accounts for stratification variables (size, subsector, state) and adjusts for differential response rates across demographic and geographic subgroups. Analytical models incorporate these weights using Stata's `svyset` command to produce population-level estimates, following Agresti & Finlay's (2014) guidance on complex survey analysis.

Comparative Advantages

While alternative datasets like the National Center for Charitable Statistics (NCCS) Core Files provide financial metrics, this survey uniquely captures:

1. Community context: ZIP code-level economic data linked to organizational outcomes

2. Revenue diversification: Herfindahl-Hirschman Index (HHI) calculated from 10 funding streams
3. COVID-19 impacts: Year-over-year donation changes (2019–2020) during a period of acute fiscal stress

This combination of financial, geographic, and temporal variables addresses critical gaps in nonprofit research, which often lacks standardized measures of community economic conditions (Bowman, 2011). The exclusion of educational and medical institutions limits generalizability to those subsectors but strengthens focus on community-based organizations facing distinct resource challenges.

Operationalization of Key Constructs

1. The dependent variable is donation revenue change, operationalized as the year-over-year percentage change in donations from 2019 to 2020 (variable: frchanges_2_1).
2. The independent variables include:
 - Urban-Rural Classification: Binary variable indicating urban core vs. rural location (mav_urban_rural_oth_cat).
 - Low-Income Community: Binary indicator based on ZIP code-level median household income (mav_ziptostate_lowinc).
3. Control variables account for organizational size (sizestrata) and sector-specific effects. Survey weights adjust for unequal selection probabilities and non-response bias (Urban Institute, 2021).

Analytical Strategy

The analysis employs a two-stage approach:

- 1. Descriptive Analysis: Weighted means and distributions establish baseline financial health profiles
- 2. Multivariate Regression: Linear models test geographic and economic effects while controlling for size:
 - Model 1: Urban-rural effects on financial indicators
 - Model 2: Low-income community impacts

Results

This section presents key findings from the analysis. Descriptive statistics reveal moderate donation growth and reserve access across nonprofits, but bivariate and regression analyses show significant disparities by geography and income level.

Descriptive Statistics Results

Table 1: Summary Statistics for Donation Revenue Change

N	Mean	Std. Dev.	Std. Error	Min	Max	Skewness	Kurtosis
1671	5.24	14.31	0.35	1	99	6.29	41.09

Table 2: Mean Donation Change by Key Variables

Group	Mean	Std. Error	95% CI
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Rural	5.21	0.59	[4.06, 6.37]
Urban	6.01	0.97	[4.10, 7.91]
Higher-Income ZIP	5.40	0.40	[4.61, 6.19]
Low-Income ZIP	4.39	0.86	[2.71, 6.07]

Table 3: Frequencies of Key Categorical Variables

Urban Status:

- Rural: 613 (67.8%)
- Urban: 291 (32.2%)

Community Income Level:

- Higher-Income ZIPs: 1,448 (89.8%)
- Low-Income ZIPs: 164 (10.2%)

Organization Size:

- <\$100,000: 291 (17.4%)
- \$100,000–\$499,999: 752 (45.0%)
- \$500,000–\$999,999: 249 (14.9%)
- \$1M–\$9.99M: 340 (20.4%)
- \$10M+: 39 (2.3%)

Interpretation

The mean donation revenue change across the sample was 5.24%, suggesting a modest average increase between 2019 and 2020. However, the high standard deviation (14.31) and maximum value of 99 indicate that the distribution is heavily skewed by outliers. This is confirmed by the

skewness (6.29) and kurtosis (41.09), which suggest a long right tail and heavy-tailed distribution.

Urban organizations had a higher mean donation change (6.01%) compared to rural ones (5.21%), although the overlapping confidence intervals imply this difference may not be statistically significant. In terms of community wealth, nonprofits in low-income ZIP codes had lower mean donation growth (4.39%) than those in higher-income ZIPs (5.40%). This supports my second hypothesis (H2), suggesting that economic context may influence fundraising outcomes. Frequency distributions show that the majority of sampled organizations are small (under \$500K), located in higher-income ZIPs, and based in non-urban areas. These descriptive patterns provide important context for interpreting subsequent regression results.

Multivariate Regression Results

Table 2. OLS Multiple Regression Models Predicting Nonprofit Donation Revenue Change

Variables	Model 1	Model 2
Urban (ref. rural)	1.099	
	(1.178)	
Low-income ZIP		-1.264
		(0.937)
Org. size: \$10M and above	-3.236*	-0.658
	(1.392)	(2.704)
Org. size: \$100K–\$499K	-1.193	-2.046+
	(1.655)	(1.169)

Org. size: \$500K–\$999K	-1.333	-0.963
	(1.905)	(1.435)
Org. size: <\$100K	-4.081**	-2.837*
	(1.531)	(1.192)
Constant	6.687***	6.976***
	(1.361)	(1.082)
N	904	1,612
R ²	0.009	0.005

Source: National Survey of Nonprofit Trends and Impacts.

Notes: +p<0.10 *p<0.05 **p<0.01 ***p<0.001. Standard errors in parentheses.

Multivariate Analysis – Urban vs. Rural (H1)

Model 1 examines the relationship between nonprofit location (urban vs. rural) and donation revenue change, while controlling for organizational size. The regression results indicate that urban nonprofits had a higher average donation revenue change than their rural counterparts (Urban $\beta = 1.099$, SE = 1.178), but this effect is not statistically significant ($p = 0.351$). Thus, there is no evidence to support the hypothesis that urban location alone predicts donation revenue growth. In contrast, organizational size does appear to matter: nonprofits with annual budgets under \$100,000 reported significantly lower donation revenue growth ($\beta = -4.081$, SE = 1.531, $p < 0.01$), and those with budgets over \$10 million also showed a significant decline ($\beta = -3.236$, SE = 1.392, $p < 0.05$). These findings suggest that both very small and very large organizations

may face unique challenges in maintaining donation levels, potentially due to capacity constraints or fluctuations in major donor support (Bowman, 2011).

Multivariate Analysis – Low-Income ZIP Code (H2)

Model 2 adds economic context by testing whether being located in a low-income ZIP code predicts changes in donation revenue. The results indicate that nonprofits in low-income communities experience a lower average donation revenue change ($\beta = -1.264$, $SE = 0.937$), although this relationship is not statistically significant ($p = 0.178$). Organizational size again emerges as a significant factor: nonprofits with budgets under \$100,000 experienced significantly reduced donation growth ($\beta = -2.837$, $SE = 1.192$, $p < 0.05$). These findings provide partial support for the idea that economic context matters, even if the direct effect of low-income ZIP status does not reach conventional levels of statistical significance (Nesbit et al., 2021).

Together, the models highlight that organizational size is a consistent and significant predictor of donation revenue change, while geographic and economic variables have weaker, non-significant effects in this sample. The results underscore the structural vulnerabilities of small nonprofits and call for a more nuanced understanding of how location and community wealth shape financial outcomes.

Discussion and Conclusion

Geographic Influence on Donation Revenue

The results challenge assumptions that urban nonprofits inherently benefit from greater access to philanthropic resources. While urban organizations had slightly higher average donation revenue changes, the difference was not statistically significant. This suggests that rural organizations are

not necessarily disadvantaged purely based on geography. Rural nonprofits may benefit from strong local networks, trust-based giving, and community engagement strategies that offset their distance from major funders (Bridgespan Group, 2022).

Economic Disadvantage as a Primary Driver

Although not statistically significant, the negative coefficient associated with operating in a low-income ZIP code indicates a potential disadvantage in donation revenue growth for these organizations. This aligns with prior research showing that nonprofits in economically distressed communities face both higher demand for services and lower local fundraising capacity. These organizations may be especially vulnerable during times of economic uncertainty, such as the COVID-19 pandemic (Urban Institute, 2021; Grønbjerg & Paarlberg, 2020).

Implications for Policy and Practice

The findings suggest the need for targeted support to strengthen the financial health of small and community-based nonprofits. Funders should consider place-based grantmaking and unrestricted, multi-year support, particularly for nonprofits serving low-income areas. Investments in fundraising training, digital tools, and strategic planning can also enhance financial resilience (Matthews, 2022; Foundation Center & CF Insights, 2018).

Moreover, government and philanthropic funders should incorporate data-driven equity frameworks when designing grant programs to ensure that funding is not disproportionately channeled toward already resource-rich organizations. Creating localized philanthropic infrastructure such as community foundations or giving circles in under-resourced areas can also help foster homegrown funding ecosystems. Policymakers might explore tax incentives or

matching fund programs specifically aimed at encouraging donations to nonprofits in low-income or rural communities. Finally, funding strategies must be attentive to the needs of organizations at both ends of the size spectrum: large nonprofits that may face inflexible revenue structures, and small nonprofits that struggle with basic fundraising infrastructure and administrative capacity. Supporting intermediaries that provide shared services—such as financial management, compliance support, or donor database systems—could help close capacity gaps and reduce overhead burdens.

Future Research

Future studies should explore these relationships using longitudinal data to assess whether disparities persist over time. Qualitative methods, such as interviews with nonprofit leaders, could offer insights into adaptive strategies and local fundraising dynamics. Additionally, examining differences by nonprofit subsector (e.g., health, education, housing) may reveal whether certain mission areas are more affected by geographic and economic disparities in donation revenue (Calabrese, 2020; Hager, 2001).

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